MIL-L-18276C
19 JUNE 1964

SUPERSEDING
MIL-L-18276B(WEP)
10 AUGUST 1961

MILITARY SPECIFICATION

LIGHTING, AIRCRAFT INTERIOR, INSTALLATION OF

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the installation of aircraft interior lighting.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-L-5057 —Lights, Instrument, Individual, General Specification for
MIL-E-5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-6781 —Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-E-7080 —Electric Equipment, Aircraft, Selection and Installation of
MIL-P-7788 —Plate, Plastic, Lighting
MIL-M-8650 —Mockups, Aircraft Construction of
MIL-C-18012 —Control Configuration and Markings (For Plastic Lighting Plates, Control Panels and Placards)
MIL-L-21095 —Light, Aircraft, Emergency (Inertia Actuated, for Fixed-Wing Aircraft)
MIL-C-25050 —Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-L-25467 —Lighting, Integral, Aircraft Instrument, General Specification for

STANDARDS

MILITARY

MIL-STD-203 —Cockpit Controls, Location and Actuation of For Fixed-Wing Aircraft
MIL-STD-250 —Cockpit Controls, Location and Actuation of For Helicopters
MIL-STD-411 —Aircrew Station Signals
MS25027 —Light Assembly, Cockpit, Fixed

AIR FORCE-NAVY AERONAUTICAL

AN3037 —Light Assembly, Cabin Dome

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained

FSC 6220

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from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 *Installation.* Interior lighting equipment shall be installed in accordance with MIL-E-7080.

3.1.2 *Mockup.* The lighting mockup shall conform to MIL-M-8650.

3.1.3 *Locations of lighting fixtures.* The locations of lighting fixtures shall be given careful consideration in order to provide satisfactory illumination, to prevent undesirable reflections in windshields, canopies and other enclosures, and to prevent glare from a direct or reflected view of lamps, lenses and surfaces. Lighting installation shall conform to MIL-STD-203 or MIL-STD-250, as applicable.

3.1.4 *Reflections.* Reflections of instruments and consoles in windshields and other enclosures shall be avoided. If necessary, a suitable nonrigid shield shall be provided around the instrument panel or console.

3.1.5 *Red light.* Red light shall conform to MIL-C-25050 for aviation red and also shall not be paler or yellower than National Bureau of Standards No. 3215 filter.

3.1.6 *Lighting power.* All lamps shall be operated from a 5- or 28-volt (V) ac or dc, or 115V ac, 400 cycles per second (cps) power source.

3.1.7 *Explosive atmosphere.* Whenever a light is installed in an area normally containing an explosive vapor, the light shall be a vaporproof light conforming to the test requirements of Procedure IV, explosive-proof (aeronautical) tests of MIL-E-5272. Lights installed in areas which may contain explosive vapors shall be equipped with covers to prevent inadvertent breakage of the lamp(s).

3.1.8 *Mechanical damage.* Whenever a light is installed so as to be subject to damage from cargo, passenger, or crew normal movement, it shall be protected by a cover to prevent breakage of the lamp(s).

3.1.9 *Control configuration and markings.* Control configuration and markings for information plates, control panels, and placards shall be chosen, arranged and marked in accordance with MIL-C-18012.

3.2 **Visual signals.** Visual signals shall conform to MIL-STD-411. Automatic dimming shall be provided for illuminated signals so that they will be dimmed when the flight instrument lighting control is in any position but "OFF". The dim intensity of warning lights shall approximate that of the flight instruments at full intensity. Dimming shall not be provided for fire warning lights.

3.2.1 *Wheels warning light.* A warning light shall be incorporated in the landing gear control handle, connected so that the light is on whenever the handle and landing gear are not in agreement.

3.2.2 *Wheels warning legend (fixed-wing aircraft).* A "WHEELS" legend indicator shall be installed and connected so that it flashes whenever: (a) the aircraft power setting is less than minimum cruise, (b) the flaps or other high-lift devices are not fully retracted, and (c) the wheels are not down and locked.

3.2.3 *Wheels warning legend (rotary-wing aircraft).* A "WHEELS" legend indicator shall be installed on helicopters equipped with retractable wheels and connected so that it flashes whenever the airspeed and collective pitch, or other combination of functions approved by the procuring activity, indicates that the helicopter is configured for landing and the wheels are not down and locked.

3.2.4 *Flash rate.* The "WHEELS" warning legend shall flash at 90, ±10 flashes per minute with an on-off ratio of approximately 2 to 1.

3.2.5 *Hook warning light.* A warning light shall be incorporated in the hook control handle, connected so that the light is on

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whenever the handle and the arresting hook are not in agreement.

3.3 Circuit breaker panels. Circuit breaker panels shall be lighted when the aircraft is designed for in-flight maintenance.

3.4 Chart board lighting. Chart boards not self-lighted shall be provided with lighting sufficient for reading maps and charts, and for navigation.

3.5 Cockpit lighting.

3.5.1 *Instrument lighting.* The instrument lighting shall conform to this specification, unless the contractor is authorized in writing by the procuring activity to install an alternate nonstandard instrument lighting system which does not conform to this specification. The request for authorization to install an alternate lighting system shall be made before the lighting mockup inspection and shall include samples, photographs, drawings, or other data to indicate its superiority over the requirements of this specification for installation in the particular aircraft under consideration. The right is reserved to require the contractor to mock up both the alternate and the standard installations.

3.5.2 *Installation of equipment.* All items of lighting equipment specified for installation shall conform to their respective drawings, specifications, or both, and shall be installed without modification, except for electrical leads.

3.5.3 *Selection of lights.* When specified by the procuring activity, as soon as possible after each instrument panel design or design change has been determined, the contractor shall furnish the procuring activity with one sample of each instrument panel, the instrument panel drawings involved, or both, for use by the procuring activity for lighting studies before making recommendations relative to the types of lights to be used. The drawings shall show the layout of each instrument panel and the relative position of the crew member's normal eye level with respect to the instrument panel under his observation.

3.5.4 *Mockup.* In order to determine the adequacy of the instrument lighting systems for each new model aircraft, or modification thereof, which would affect the design of the instrument panel layout or the lighting system, a complete mockup of the instrument lighting system involved shall be prepared by the aircraft contractor for inspection and approval by the procuring activity. The mockup shall be provided with the actual lighting equipment and instruments or models of instruments to be illuminated, insofar as practicable, which will afford a reasonable indication of the effectiveness of the lighting involved. Cardboard templates of instrument dials will not be satisfactory for this mockup.

3.5.5 *Instrument panels.* The instruments shall be provided with primary and secondary lighting systems.

3.5.5.1 *Primary lighting system.* The primary instrument lighting system shall employ lights conforming to MIL–L–5057 and shall provide red light in Naval aircraft and white light in Army aircraft, except that, for integrally lighted instruments, the lighting shall be in accordance with MIL–L–25467. The light(s) for each instrument shall be installed in the number, location(s), and orientation(s), when applicable, for which the light is qualified, as specified in MIL–L–5057, for lighting that instrument case. For lights having lead wires without sockets, enough slack shall be provided in the electrical lead of each instrument light in order that between 3½ to 4 inches of the lead can be readily pulled through the instrument panel upon removal of the light from the instrument or panel.

3.5.5.2 *Secondary lighting system.* The secondary instrument lighting system shall be provided with lights in accordance with MS25027, except that the lights shall provide white light in Army aircraft. Sufficient lights shall be employed to provide adequate illumination subject to the mockup approval. Where practicable, these lights shall be installed under the glare shield of the instru-

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ment panel, as far away from the instrument panel as practicable.

3.5.6 *Console lighting.* The consoles shall be provided with primary and secondary lighting systems.

3.5.6.1 *Primary lighting system.* The primary console lighting system shall consist of integrally lighted information plates conforming to MIL-P-7788. The plates shall be black or gray in Naval aircraft and gray in Army aircraft.

3.5.6.2 *Secondary lighting systems.* The secondary console lighting system shall provide visibility of switch handles, knobs, and controls. The fixtures employed shall be as approved by the procuring activity, and shall provide red light in Naval aircraft and white light in Army aircraft. Sufficient fixtures shall be employed to adequately light the switch handles, knobs, and controls.

3.5.7 *Emergency lighting system.* An emergency lighting system shall be provided for use in the event of failure of the primary and secondary instrument lighting systems or the primary console lighting system. If the secondary console lighting fixtures can be removed from their bases for hand holding, these lights may be employed for the emergency lights. In Naval aircraft, the emergency lights shall be readily changeable from red to white light while in use. In Army aircraft, the emergency lights shall provide white light. At least one of the emergency lights shall be directable by hand or from its base to within 2 feet of all parts of the cockpit.

3.5.8 *High altitude and nuclear weapon lighting system.* Aircraft which have a service ceiling higher than 40,000 feet or are designed to deliver nuclear weapons for air or surface burst, shall be provided with an additional system for illuminating the instrument panels and consoles with white light. The illumination provided shall be a minimum of 150 foot-candles for the primary flight instruments and a minimum of 75 foot-candles for other instruments and controls in the cockpit. Continuously variable manual control shall be provided. In addition, in aircraft equipped with an automatic thermal protective closure system, the light(s) shall be automatically turned on to full bright when the closure is closed. A momentary contact reset switch shall be provided to regain manual control.

3.5.9 *Antidazzle lighting system.* Except for fighter and interceptor aircraft, and aircraft which have high altitude and nuclear weapon lighting systems in accordance with 3.5.8, one or more lights conforming to AN3037-8 shall be installed in each cockpit for emergency maintenance and flight during lightning storms. The lights shall flood the instrument panel, controls, and floor of the cockpit with at least 10 foot-candles.

3.5.10 *Controls.* Standard control panels in accordance with MIL-C-6781 shall be provided for control of cockpit lights.

3.5.10.1 *Locations.* Each crew station shall be provided with a lighting control panel, convenient for operation by that crewman, for controlling all lights which are usually used by that crewman only. For lights which are usually used by two crewmen, one lighting control panel convenient for operation by both of those crewmen shall be provided. Where required for satisfactory control because of the great number of instruments, consoles lit, or because some are overhead and some below eye level, etc., two lighting control panels shall be provided. Each lighting control panel which controls lighting of both flight and nonflight instruments of which the flight instruments constitute a total of more than four instruments, or more than one quarter of the total number of instruments, or both, shall have separate controls for the primary lighting of the flight and nonflight instruments. All other lighting control panels which control lighting of both flight and nonflight instruments shall have one control for the primary lighting of both the flight and the nonflight instruments.

3.5.10.2 *Functions.* Each lighting control panel shall include, as applicable, one or two instrument primary lighting system controls

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marked "INST", or "FLT INST" and "NON-FLT INST", as applicable, one console primary lighting system control marked "CONSOLE", and one instrument and console secondary lighting system control marked "FLOOD".

3.5.10.3 *Operation.* Each control shall operate as follows.

3.5.10.3.1 *Primary lighting controls.* Each primary lighting control shall be a rotary control, operated by a standard control knob. When the knob is in its extreme counterclockwise position, the lights shall be off. Rotating the knob clockwise from its "OFF" position shall first turn the lights on with between zero and one-seventh normal potential applied to the lamps, and then shall increase the potential applied to the lamps, either continuously or in at least seven steps, until when the knob is in its extreme clockwise position, normal potential shall be applied to the lamps.

3.5.10.3.2 *Secondary lighting controls.* Each secondary lighting control shall be a three-position toggle switch installed so that its handle is moved upward and downward to operate it. The switch shall be connected to the source of power through a means operated by the console primary lighting system control, such that when the console primary lights are off, the console and instrument secondary lights will be off. When the console primary lights are on, the console secondary lights will be on, with their brightness controlled by the secondary lighting control. The instrument secondary lights will be turned on and off by the secondary lighting control, as follows:

(a) When the control is in its center position, marked "DIM", one-third normal potential shall be applied to the console secondary lights and no potential shall be applied to the instrument secondary lights.

(b) When the control is in its lower position, marked "MED", two-thirds normal potential shall be applied to the console secondary lights and no potential shall be applied to the instrument secondary lights.

(c) When the control is in its upper position, marked "BRT", normal potential shall be applied to the console and instrument secondary lights.

3.5.10.3.3 *Power sources.* The console and instrument secondary lighting systems and the emergency lighting system shall be connected to the essential bus of the aircraft electric system. The other lighting systems shall be connected to the primary bus. Each lighting system shall be provided with an individual fuse, circuit breaker, or other electric protective device.

3.5.11 *Check list and radio call plates.* Each check list and radio call plate shall be an integrally lighted information plate in accordance with MIL-P-7788, the lighting of which shall be controlled by the same device that controls the primary lighting system of the nearest instruments or consoles, as applicable, to the plate. The plates shall be black in Navy aircraft and gray in Army aircraft.

3.6 Lighting in compartments other than cockpits.

3.6.1 *Instruments and consoles.* In compartments in Navy aircraft where dark adaptation is required, or from which stray white light could affect dark adaptation of crewmen in other compartments requiring it, and in all compartments in Army aircraft, instruments and consoles shall be provided with primary and secondary lighting systems and controls in accordance with 3.5. For compartments in Naval aircraft where dark adaptation is not required, and from which stray white light cannot affect dark adaptation of crewmen in other compartments requiring it (except areas around video displays which are lighted in accordance with 3.6.2), instruments and consoles shall be provided with primary and secondary lighting systems and controls in accordance with 3.5.

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All lighting systems, except the primary instrument lighting system, shall provide white light. Integrally lighted information plates shall be gray. The secondary instrument lighting system shall be controlled by a separate rotary control marked "INST FLOOD" as specified in 3.5.10.3.1, instead of being controlled by the same toggle switch that controls the console secondary lighting system, marked "CONSOLE FLOOD", and the console primary lighting system control.

3.6.2 *Video displays.* The area surrounding a video display in a compartment where dark adaptation is not required shall be illuminated with polarized low-level white light. The video display shall be provided with a polarized screen to prevent the area illuminations from reducing video contrast on the display. The light shall be placed so that specular reflections are not visible to the operator. The lighting shall not create glare at other stations and shall not interfere with functions required at any other station. Consoles around the video display shall be lighted by integrally lighted information plates conforming to MIL-P-7788.

3.6.3 *Other areas.* Where not lit in accordance with 3.6.1 or 3.6.2, areas where dark adaptation is required, or from which stray white light could affect dark adaptation of crewmen in other areas requiring it, shall be lit with red light in Naval aircraft. White light shall be used in Naval aircraft in areas where dark adptation is not required, and from which stray white light cannot affect dark adaptation of crewmen in other areas requiring it, and in Army aircraft. The illumination required with red lighting shall be as determined at the mockup inspection. The illumination with white lighting shall be at least the following:

Area	*Minimum illumination with white light (in foot-candles)*
Galley (on working surface)	10
Mess Room (on table)	10
Navigator's tables and other locations where reading and writing is required	10
Officers' and crew's quarters (5 feet above floor)	10
Radio and electronic equipment controls and components requiring adjustment	6
Compartments for auxiliary power plant, anchor gear, etc. (2 feet above floor)	6
Wash room (5 feet above floor)	4
Gunners' position	3
Bombardier's compartment (on floor)	3
Storage space (on floor)	1
Passageways (on floor)	0.2

3.7 Emergency exit lighting. All aircraft designed to carry passengers or crew in any compartment other than the cockpit shall be provided with emergency exit lights conforming to Civil Air Regulations. The lights shall illuminate all emergency exit markings. The lights shall conform to MIL-L-21095 and shall be operable manually and be removable to allow them to be used as portable lights. Immersion-operated switches shall be provided to operate the lights in rotary-wing aircraft.

4. QUALITY ASSURANCE PROVISIONS

4.1 Each instrument lighting installation or operation shall be carefully examined to determine conformance with all the requirements of this specification.

5. PREPARATION FOR DELIVERY

Not applicable.

6. NOTES

6.1 Intended use. This specification is intended to be used for the installation of aircraft interior lighting.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-L-18276C Lighting, Aircraft Interior, Installation of

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity)	DATE

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